

LCDSirPlus 0.3.0 Instruction Manual

Draft / Unreleased: 0.3.0 release acceptance is pending. Installer and portable-package procedures apply only if those artifacts are published.

LCDSirPlus is a Windows 11 x64 dashboard for the Logitech G13 160x43 monochrome LCD. It runs as a standard user, provides a virtual preview, and uses explicit `N/A`, stale, offline, or disabled states when optional telemetry is unavailable.

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Requirements and dependencies

Required:

- Windows 11 x64 and a standard, unelevated user account.
- A Logitech G13 only for the physical LCD and buttons. The built-in virtual preview works without one.
- A writable configuration location: installed copies use `%LOCALAPPDATA%\LCDSirPlus\Config\lcdsirplus.txt`; portable/development copies use `lcdsirplus.txt` beside `LCDSirPlus.exe`.

The main application is one native executable. Published release packages are designed to include the official signed PresentMon v2.5.1 console for frame timing; LCDSirPlus does not install a runtime, service, driver, listener, browser extension, PresentMon MSI/service, or elevated component. LCDSirPlus packages are not code signed, so verify their SHA-256 checksums before use. Package-level dependency notices are in `THIRD_PARTY_LICENSES.txt`; PresentMon's notices remain under `licenses/PresentMon/`.

Optional features have separate dependencies:

Feature	Dependency
G13 through Logitech software	Installed, running Logitech Gaming Software (<code>LCore.exe</code>) with its trusted LCD SDK. <code>auto</code> otherwise attempts direct HID, which also requires Logitech LampArray service to release the G13.

Feature	Dependency
GPU load, VRAM, temperature	NVIDIA or AMD display driver exposing NVAPI or ADLX.
CPU temperature	Preferred: user-managed HWiNFO Sensors with Shared Memory Support and an exact configured label pair. Optional fallback: LibreHardwareMonitor with its loopback web server explicitly enabled by the user.
FPS, frame time, game name, and sessions	Bundled official signed PresentMon v2.5.1 console, <code>PresentMon.exe</code> ; plain Cargo output does not bundle it.
Headset battery	Supported SteelSeries Arctis 1, 7X/7P, 9, Pro Wireless, 7+/7P+, Nova 3/5/7, or GameBuds wireless USB receiver family.
Controller battery	Connected XInput controller.
Discord speaker overlay	Discord Desktop, a Discord developer application, tester access when required, and one authorization per Discord account.
Ping, jitter, and packet loss	An explicitly enabled ICMP/TCP probe to one configured IP literal.

Install or run portable

Verify a release

Place the checksum file beside the downloaded artifacts and compare the published values with the local files:

```
Get-Content .\LCDSirPlus-0.3.0-SHA256SUMS.txt
Get-FileHash .\LCDSirPlus-0.3.0-* -Algorithm SHA256
```

Each ZIP also contains a sorted `PACKAGE-MANIFEST.txt` listing every member's SHA-256 and byte size. The setup EXE is covered by the outer checksum file.

Installer package

If published, run `LCDSirPlus-0.3.0-win-x64-setup.exe`. Current-user mode installs to `%LOCALAPPDATA%\Programs\LCDSirPlus`. The mode dialog can select an elevated all-users install to `%ProgramFiles%\LCDSirPlus`. Start Menu and per-user sign-in startup tasks are checked by default; the desktop shortcut is unchecked. The startup integration is an interactive, limited scheduled task, not an HKCU Run value or service. Its stable name includes the owning Windows account SID, so different accounts do not share one task identity. Setup installs a default configuration template. On first installed launch, the application validates and atomically seeds `%LOCALAPPDATA%\LCDSirPlus\Config\lcdsirplus.txt` only if that user file is absent; update and repair do not replace it.

Portable package

Extract the portable ZIP to a local directory and run `LCDSirPlus.exe` there. Keep `lcdsirplus.txt` beside it. Portable use makes no installer-owned shortcut. The package includes `modules.md` as a simple slot-value guide.

First start

1. Validate the shipped configuration:

```
.\LCDSirPlus.exe --validate-config
```

2. Start with a forced preview so the dashboard can be checked without taking the G13 display:

```
.\LCDSirPlus.exe --preview
```

Preview is a long-running normal application mode and owns the per-session LCDSirPlus mutex. Close it before a nonvirtual hardware test. The tray icon remains available whenever the application runs; the preview title-bar X exits the application rather than merely hiding the window.

3. If using a physical G13, inspect it without writing to the LCD:

```
.\LCDSirPlus.exe --hardware-discover
```

4. Exit Logitech Gaming Software normally before a direct-HID test, then run:

```
.\LCDSirPlus.exe --hardware-test --backend hid --duration-secs 60
```

5. For the SDK path, start Logitech Gaming Software normally and run:

```
.\LCDSirPlus.exe --hardware-test --backend sdk --duration-secs 60
```

Watch the physical LCD and press all four LCD buttons during hardware tests. Command success proves transport submission, not physical display quality.

Normal `logitech_backend auto` selects the trusted Logitech SDK while exact process `LCore.exe` owns the G13 and otherwise attempts direct HID. Direct HID also refuses Logitech LampArray when that service owns the device. Explicit `sdk` and `hid` modes do not cross-fallback. LCDSirPlus never stops Logitech software for you.

Slots and buttons

The lower row has four slots aligned with the four physical LCD buttons. Each `slot_N` configuration line is an ordered list. A short press cycles only its matching slot and wraps at the end:

```
slot_0  HEADSET_BATTERY CPU_TEMP CONTROLLER_BATTERY
slot_1  FPS_CURRENT FPS_1LOW FRAME_TIME SESSION_TIME
slot_2  PROC_HANG GPU_TEMP PING JITTER AUDIO
slot_3  THERMALS PACKET_LOSS MIC_STATUS SESSION_SUMMARY PROVIDER_STATUS
```

In the preview, left-click a slot to cycle forward and right-click it to cycle backward. The tray icon always exists while the application runs: left-click it to toggle the preview, or right-click it for Toggle Preview and Exit. Closing the preview with its title-bar X exits LCDSirPlus.

Physical button 4 or preview slot 4 first acknowledges the highest active alert before performing its normal slot action. When the guarded hung-window UI is active, its selected slot's physical button has special hold/release behavior described below.

Every graph shows exactly the trailing 30 seconds. Temperature and memory warnings are enabled independently. With temperature warnings enabled and `warning 1`, only the selected CPU/GPU temperature pane inverts in alternating 100 ms phases at its configured maximum; CPU/GPU temperature alert episodes do not use full-screen overlays. With `warning 0`, those temperature overlays remain. Memory and headset alerts are unaffected by this presentation setting.

Main displays

The compact date/time header and fixed four-button bottom row stay the same. Select one of three fixed built-in metric layouts:

- `main_display 1` (default): original CPU/RAM and GPU/VRAM halves.
- `main_display 2`: CPU/RAM, GPU/VRAM, and OUT/IN thirds.
- `main_display 3`: original CPU/RAM on the left and NET IN/NET OUT on the right.

The value is an integer and only `1`, `2`, or `3` is valid. Saving a valid change applies it on hot reload. The OUT/IN bars show current throughput using `network_graph_ceiling_mbps`; with its default `1000` Mbps ceiling, 1 Gbps fills a bar. Layout 1 has no network bars.

Configuration and preview

The text format uses whitespace-separated keys and values, `#` comments, and quoted values for embedded spaces. The executable watches the primary file and hot-reloads valid changes. Tokenization and parse errors identify their source file and line. Final range and cross-field validation errors may report line `0` because no exact source line is retained for those checks. The last valid configuration remains active after any invalid edit.

Installed copies default to `%LOCALAPPDATA%\LCDSirPlus\Config\lcdsirplus.txt`; portable/development copies use the adjacent file. Cargo copies the canonical root configuration beside debug and release executables. An explicit `--config PATH` has precedence for normal runtime, validation, hardware tests, and Discord authorization/credential removal, as detailed under Command line.

Validate before or after editing:

```
.\LCDSirPlus.exe --validate-config
.\LCDSirPlus.exe --validate-config --config C:\path\lcdsirplus.txt
```

The complete key, default, and range reference is in CONFIGURATION.md. Important initial settings are:

Key	Purpose
<code>main_display 1</code>	Select fixed built-in layout 1, 2, or 3; applies on valid hot reload.
<code>temperature_warning_enabled 1</code>	Enable CPU/GPU temperature episodes and their selected presentation.
<code>memory_warning_enabled 1</code>	Enable RAM/VRAM capacity episodes independently.
<code>memory_warning 100</code>	RAM warning percentage, inclusive at <code>>=</code> ; <code>0</code> is the compatibility disable value.
<code>vmem_warning 100</code>	VRAM warning percentage, inclusive at <code>>=</code> ; <code>0</code> is the compatibility disable value.
<code>warning 1</code>	With temperature warnings enabled, flash selected panes and suppress temperature full-screen overlays; <code>0</code> retains overlays.
<code>preview_mode auto</code>	Show preview when a physical HID/SDK backend is unavailable. <code>always</code> starts it visible; <code>never</code> starts it hidden.
<code>preview_scale 4</code>	Preview scale, accepted range 1 through 10.
<code>start_minimized 0</code>	Suppress automatic preview display when set to 1.
<code>start_at_login 0</code>	In portable mode, own the current user's exact LCDSirPlus Run value when set to 1. Installed startup is selected in setup instead. Safe mode never changes it.
<code>logitech_backend auto</code>	Select trusted SDK with LCore, otherwise direct HID. <code>virtual</code> is preview-only.

Key	Purpose
<code>hang_enabled 0</code>	Keep hung-window detection and destructive termination disabled until explicit opt-in and physical disposable-child qualification.
<code>safe_mode 0</code>	Disable providers, Discord, probes, startup mutation, and the destructive hung action.

`preview_scale`, `log_level`, `log_max_bytes`, and `log_backups` are validated during hot reload but take effect only after restart. `date_format` and `time_format` are accepted but currently do not change the shared header. `start_minimized` describes automatic-start preview behavior.

The two warning category switches apply on valid hot reload. Disabling temperature warnings prevents CPU/GPU episodes and pane flashing without affecting memory or headset alerts. Disabling memory warnings prevents RAM/VRAM episodes without affecting temperature or headset alerts. Stale and unavailable readings never trigger. Use the category switch instead of zero thresholds for new configurations; installer updates preserve existing explicit thresholds and do not rewrite them. Existing episodes remain through unknown or stale readings. A first current valid recovery removes severity-2 warnings immediately and starts `critical_alert_linger_ms` only for severity-3 episodes. Disabling a warning category clears its episodes; disabling the headset provider clears its episode.

An include file can hold machine-local overrides in source or portable use:

```
include lcdsirplus.local.txt
```

Relative includes cannot use `..`, absolute paths, or environment expansion. Later included values override earlier files; duplicate keys in one file are rejected. Installed relative includes belong beside the user configuration under `%LOCALAPPDATA%\LCDSirPlus\Config`, which setup preserves.

Providers

Built-in Windows sources

- Local date/time uses the shared fixed Windows date/time header layout.
- CPU load uses native scheduler accounting and CCD topology detection.
- RAM percentage uses `GlobalMemoryStatusEx`.
- Network throughput uses native interface counters, excluding loopback and interfaces that are down.
- Audio volume and microphone mute use the default Windows Core Audio endpoints.
- Controller battery uses `XInput`.
- Aggregate disk I/O and approximate page-read pressure use Windows PDH.
- Established connections use native IPv4 and IPv6 TCP tables.
- AC/battery state uses `GetSystemPowerStatus`.

These sources require no downloaded helper. Native Win32 and installed vendor APIs always take priority. HWiNFO or exact/automatic LibreHardwareMonitor selection is used only where no safe native source exists. LCDSirPlus never probes raw MSRs, SMBus, EC, or Super-I/O registers.

GPU

`gpu_provider auto` tries the vendor-installed NVIDIA NVAPI provider and then AMD ADLX. `nvapi` or `adlx` selects one explicitly; `off` disables both. Native GPU telemetry can provide load, VRAM used/total, and temperature. LHM can only fall back for temperature; it does not provide LCDSirPlus GPU load or VRAM.

CPU temperature: HWiNFO and LibreHardwareMonitor

Current Windows has no generic CPU package-temperature API. `CPU_TEMP` can be `N/A` because no source is configured, HWiNFO shared memory/LHM is stopped, the configured label pair or SensorId is missing or ambiguous, a value has the wrong type/unit, the data is stale, or the shared-memory ABI is unsupported.

The preferred fallback is user-managed HWiNFO:

1. Start HWiNFO in **Sensors-only** mode.
2. Enable **Shared Memory Support** in HWiNFO settings.
3. Run `LCDSirPlus.exe --list-hwinfo-sensors`.
4. Find the intended CPU package/control temperature line.
5. Copy its exact case-sensitive `sensor=` and `reading=` original labels into:

```
hwinfo_cpu_temp_sensor "<exact sensor original label>"
hwinfo_cpu_temp_reading "<exact temperature original label>"
```

6. Validate configuration and start LCDSirPlus.

Both keys must be configured together. HWiNFO64 Free limits shared-memory monitoring to 12 hours. For unattended use, use HWiNFO32 or an appropriately licensed HWiNFO Pro edition according to HWiNFO's vendor terms. LCDSirPlus does not bundle, start, configure, restart, or license HWiNFO. Access is read-only, and configured original labels are exact and case-sensitive.

LibreHardwareMonitor is an optional fallback restricted to loopback HTTP. The user must explicitly run it and safely enable its web server; LCDSirPlus does not enable or manage that server and does not request a firewall exception. At the default address, use:

```
lhm_mode auto
lhm_url auto
```

`auto` resolves to `http://127.0.0.1:8085/data.json`. If automatic temperature selection is wrong, copy stable SensorIds from that live loopback `data.json` response, not from an LCDSirPlus diagnostics bundle, and set the overrides documented in `CONFIGURATION.md`. LHM is user-managed and not bundled; its absence is an acceptable unavailable state. The accepted `CPU_CACHE_TEMP` and `CPU_FREQ_TEMP` options are also currently unavailable because no runtime provider publishes the required per-domain temperatures.

PresentMon

Published release packages are designed to bundle Intel's official signed PresentMon v2.5.1 console as `PresentMon.exe`. Default `presentmon_path auto` uses that colocated file; plain Cargo output does not bundle it. Advanced users can select a valid console on a trusted fixed local drive explicitly. Default foreground targeting follows the current foreground process. Process-name mode requires `presentmon_process_name`.

LCDSirPlus launches and owns PresentMon only while an enabled target frame capture is active and stops its own child on target/config change or shutdown. It provides FPS, low FPS, frame time, game name, session duration, and stutter count only; it does not provide CPU/GPU hardware telemetry. No PresentMon service, MSI, GUI, or API/SDK is bundled or installed. Data becomes stale after five seconds without frames and expires after another five.

If capture cannot start, add the user to Windows' **Performance Log Users** group and sign out/in, or test elevation if local ETW policy requires it. Elevation is troubleshooting, not a normal requirement. PresentMon is MIT licensed; its

exact notices are in `licenses/PresentMon/LICENSE.txt` and `licenses/PresentMon/THIRD_PARTY.txt`, and upstream is [GameTechDev/PresentMon](#).

The provider is implemented, software-tested, pinned, and remains enabled by default because it is a required feature. Live capture against an actively presenting game is not yet release-qualified and is deferred while this PC's memory is occupied by the local LLM. Release remains pending that live gate.

Headset, controller, and audio

The headset provider supports an explicit set of SteelSeries Arctis 1, 7X/7P, 9, Pro Wireless, 7+/7P+, Nova 3/5/7 Wireless, and GameBuds USB receiver families. It does not generically support every SteelSeries, Bluetooth, or wired device. Connect the wireless receiver through USB; Bluetooth alone does not expose the supported control interface. SteelSeries GG may coexist, but LCDSirPlus reads the receiver directly and does not require GG.

Battery values follow the hardware: direct or scaled percentages on some models and 0/25/50/75/100% bands on others. GameBuds shows the lower battery percentage of the active earbuds, not docked earbuds or the case. The title changes to `CHARGING` only when that profile supplies a charging state; the value remains a complete percentage. Exact receiver PIDs and profile details are in `CONFIGURATION.md`.

XInput battery levels are the platform's coarse values. Core Audio reads the current default multimedia endpoints. Disable an unneeded provider with its `*_enabled 0` setting.

`PROVIDER_STATUS` summarizes providers currently tracked by the runtime. `OK` means none of those tracked providers is marked failed; it does not mean every optional provider is installed or enabled. Safe mode publishes only its healthy `safe-mode` state, so the slot renders `OK` rather than counting disabled Discord, hung-window, or other providers.

Network rates and quality

Throughput and quality probing are independent:

- `NET_IN`, `NET_OUT`, and `NET_BOTH` use local interface counters and do not send network traffic. Numeric values are instantaneous decimal SI bit rates.
- `NET_IN_GRAPH` and `NET_OUT_GRAPH` show exactly the trailing 30 seconds in one-second bins.
- `NET_GRAPH` shows ingress above and egress below its center line for the same 30-second history.
- `PING`, `JITTER`, and `PACKET_LOSS` require the optional active probe, which is off by default.

All network graph variants and the current OUT/IN bars in main displays 2 and 3 share `network_graph_ceiling_mbps`. Its default is `1000` and accepted range is `1..100000` Mbps. At the default, 1 Gbps fills a bar. The ceiling does not change numeric readings. Example for a 2.5 Gbps link:

```
network_graph_ceiling_mbps 2500
```

To enable a bounded quality probe:

```
network_probe_enabled 1
network_probe_method auto
network_probe_target 1.1.1.1
network_probe_interval_ms 1000
network_probe_timeout_ms 1500
network_probe_window 30
```

The target must be one IP literal, optionally with a TCP port. Hostnames are rejected. `icmp` accepts IPv4 without a port. `tcp` accepts IPv4 or IPv6 and defaults to port 443 when omitted. `auto` tries IPv4 ICMP then TCP within one overall timeout; IPv6 goes directly to TCP. Safe mode and a disabled probe send no probe traffic.

Discord setup

Discord integration uses the trust-checked local Discord Desktop named pipe for voice state and bounded HTTPS only for token exchange/refresh. It opens no browser, callback listener, or portal redirect.

Running Discord alone is insufficient. `discord_client_id` must be the public numeric Application ID/OAuth client ID from a Discord Developer Portal application; it is not a secret. If the ID is missing, Discord is unconfigured and no IPC connection is attempted, so this is not a connection failure. LCDSirPlus reads voice state only: it publishes no Rich Presence and requires no image assets.

The integration is implemented and software-tested, but its live voice/OAuth workflow is not yet release-qualified. Each user creates and registers their own Discord application, and tokens remain current-user DPAPI-protected local data. Release remains pending this gate unless it is explicitly deferred.

1. Create your own Discord developer application, obtain its OAuth client secret, and add the account as an application tester if Discord requires it.
2. Put the numeric application ID in configuration:

```
discord_enabled 1
discord_client_id 123456789012345678
```

3. Start Discord Desktop and activate the account to authorize.
4. In PowerShell, enter the client secret through a masked prompt and run authorization. The `finally` block clears the temporary environment value even if authorization fails:

```
try {
    $env:LCDSIRPLUS_DISCORD_CLIENT_SECRET = Read-Host 'Discord client secret' -
MaskInput
    .\LCDSirPlus.exe --discord-authorize
} finally {
    Remove-Item Env:\LCDSIRPLUS_DISCORD_CLIENT_SECRET -ErrorAction SilentlyContinue
}
```

LCDSirPlus exchanges the RPC authorization code through Discord's generic OAuth token endpoint, which requires the client secret. Enabling **Public Client** does not make this implemented flow secretless; Discord's no-secret flow is specific to Social SDK `GetToken`, which LCDSirPlus does not implement. Never put a secret or token in source, configuration, command arguments, logs, screenshots, issues, or chat. Each Discord Account Switcher account must be active and authorized once. Credentials, including the secret retained for refresh, are encrypted for the current Windows user with DPAPI and stored under that Discord user ID in `%LOCALAPPDATA%\LCDSirPlus`.

Discord Desktop must continue running as the same Windows user and in the same interactive session. LCDSirPlus accepts only a named-pipe server whose process, session, fixed-local executable, Authenticode signature, and Discord publisher identity pass its checks.

Optional overlay controls include `discord_linger_ms`, `discord_max_speakers`, `discord_show_self`, and `discord_show_channel`. To remove all LCDSirPlus Discord credential files:

```
.\LCDSirPlus.exe --discord-clear-token
```

Revoke the application separately under Discord **User Settings > Authorized Apps** when authorization should also be invalidated remotely. Local credential removal and remote revocation are separate operations; do both after suspected credential exposure.

Safe mode and hung-window action

Start safe mode temporarily:

```
.\LCDSirPlus.exe --safe-mode
```

or set `safe_mode 1`. Safe mode disables optional telemetry providers, Discord access, network probes, hung-target binding/termination, and startup registration changes. Clock, native CPU load, native RAM percentage, preview, slot cycling, and alert acknowledgement remain harmless local functions.

The hung-window detector and destructive action are disabled by default. Set `hang_enabled 1` only to opt in. Until then, `PROC_HANG` remains in the shipped slot with the same button position and cannot bind or terminate a target. The provider reports disabled/unavailable while the existing no-target pane shows the next configured token without changing selection.

When enabled, the detector can expose repeatedly unresponsive windows. `PROC_HANG` may occur only once across all slot lists. The token's selected slot owns the matching physical button at runtime; legacy `hang_button` remains accepted but does not choose that binding. With no target, the next active token or `CLEAR` renders without changing selection. Button-down binds one exact displayed target. A short release toggles details, then advances to the next target when details are already shown; it does not terminate. A continuous hold automatically requests termination at `hang_hold_ms` after revalidation; release afterward only resets and cannot act again. If the app loop resumes only after twice the configured hold, clamped to 3..15 seconds, the hold is refused as stale. Recovery, target or selection change, device loss, provider failure, reload, or safe mode cancels the action. Safe mode leaves ordinary short-release navigation available because no target is bound. Termination can lose unsaved work. Test only with the disposable harness in `HARDWARE-ACCEPTANCE.md`, never valuable work. The action remains unqualified until that disposable-child physical-button gate passes; software tests do not qualify termination for release.

`BOTTLENECK` uses sustained configurable CPU, GPU, memory, and disk heuristics. When its current result is `NONE`, it dynamically renders the next active token, as does an inactive `PROC_HANG`; neither fall-through changes the selected token or cycle index.

Command line and exit codes

```
LCDSirPlus.exe [--config PATH] [COMMAND]
(none)           Run the GUI/tray dashboard
--validate-config Validate configuration and exit
--list-hwinfo-sensors List exact usable HWiNFO labels and exit
--preview        Force preview for the normal long-running runtime
--hardware-test  Run the deterministic G13 test sequence
--hardware-discover List compatible G13 HID discovery results without writes
--diagnostics    Write a bounded offline diagnostics ZIP and exit
--discord-authorize Authorize the active Discord Desktop account
--discord-clear-token Remove local LCDSirPlus Discord credentials
--backend MODE   auto, sdk, hid, or virtual; hardware test only
--duration-secs N 1..3600; hardware test only
--safe-mode       Disable providers and destructive actions in normal runtime
--diagnostic-dir PATH Select runtime/test logs or diagnostics output
--version, -v     Print version
--help, -h       Print help
```

`--config` applies to normal runtime, validation, hardware tests, Discord authorization, and Discord credential removal. Diagnostics deliberately ignores it; hardware discovery, HWiNFO listing, help, and version do not read it. An explicit path takes precedence. Without one, installed copies use `%LOCALAPPDATA%\LCDSirPlus\Config\lcdsirplus.txt`, while portable/development copies use adjacent `lcdsirplus.txt`; first installed launch seeds a missing user file from the installer template. `--diagnostic-dir` applies to normal runtime, hardware-test logs, and diagnostics output. `--preview` affects normal runtime. Diagnostics records the CLI safe-mode state but still starts no providers; other one-shot commands do not need safe mode. Only one command may be selected, and `--backend` / `--duration-secs` require `--hardware-test`.

Exit code	Meaning
0	Success.
1	Operational failure such as logging, diagnostics, HWiNFO, or Discord failure.
2	Invalid arguments or configuration.
3	No compatible G13 candidate, required owner, or selected hardware transport.
4	One or more hardware-test submission steps failed.
5	The single-instance mutex is already owned or could not be acquired.

Troubleshooting

Configuration will not start or reload

Run `--validate-config --config PATH`. For tokenization or parse errors, correct the reported source file and line. Final range and cross-field validation errors may report line 0; inspect the named settings because no exact source line was retained. Unknown slot tokens, duplicate keys in one file, invalid ranges, include cycles, and unsafe include paths are rejected. A failed hot reload leaves the previous valid configuration active.

G13 is not detected

Run `--hardware-discover`. For direct HID, exit Logitech Gaming Software, confirm `LCore.exe` is absent, and stop Logitech LampArray service if it owns the G13. For SDK mode, start the installed Logitech Gaming Software

normally. Do not copy SDK DLLs or kill unrelated G HUB services. Use `--preview` or `logitech_backend virtual` to isolate rendering from hardware.

Preview does not appear

Use `--preview`, set `preview_mode always`, and ensure `start_minimized 0`. Left-click the tray icon to toggle it. In `auto`, the preview normally hides while a physical HID/SDK backend is connected.

A metric says N/A, STALE, OFFLINE, or DOWN

Check the slot table below for its source. Confirm the matching provider is enabled and its hardware/program exists. Review the log for provider detail. `N/A` means no valid current value, `STALE` means a prior value stopped updating, and offline/disabled messages identify expected provider states.

For `HEADSET_BATTERY`, `NO_DONGLE` means the enabled provider could not read an exact supported receiver/control interface; it can also follow a USB open or query failure. `OFFLINE` means a supported receiver answered but the headset or active earbuds are off/disconnected. `STALE` preserves the last valid battery after updates stop, and safe mode shows `DISABLED`. Find `VID_1038&PID_XXXX` in Device Manager's Hardware Ids and compare `XXXX` with the 32-PID table in `CONFIGURATION.md`. If listed, reconnect the USB receiver and inspect the log for interface, usage, report-length, open, or timeout detail. A matching marketing name without a matching PID and control collection is not sufficient.

Known current limitations:

- `CPU_CACHE_TEMP` and `CPU_FREQ_TEMP` are accepted but have no runtime temperature producer and render `N/A`.

CPU temperature remains N/A

Run `--list-hwinfo-sensors`. If it fails, confirm HWiNFO Sensors is running and **Shared Memory Support** is enabled. If it succeeds, copy one exact original sensor/reading label pair; do not shorten or translate labels. Check the log for inactive mapping, stale data, unsupported ABI, incomplete pair, ambiguity, or invalid unit/value. HWiNFO64 Free shared-memory monitoring stops after 12 hours; restart it for interactive use or use HWiNFO32/appropriately licensed Pro for unattended use under vendor terms.

For LHM fallback, confirm the user explicitly started LHM, enabled its web server, and kept `lhm_ur1` on loopback. LCDSirPlus cannot turn that server on.

PresentMon capture remains N/A

Confirm `PresentMon.exe` remains beside `LCDSirPlus.exe`, `presentmon_enabled 1`, targeting is not disabled, and the foreground/explicit process is actively presenting frames. Do not install the PresentMon service or MSI for LCDSirPlus. If logs show ETW/capture access failure, add the user to **Performance Log Users** and sign out/in; elevation can be tested only to diagnose restrictive local policy. PresentMon supplies no hardware telemetry, so it cannot fix CPU/GPU temperature or load-source failures.

Ping/jitter/loss remain unavailable

The probe is disabled by default. Set `network_probe_enabled 1`, use a valid IP literal and method, validate the file, and check firewall/network policy. Jitter needs at least two successful samples. Safe mode always disables probing.

Discord overlay does not appear

An empty `discord_client_id` is an unconfigured state, not a failed connection; LCDSirPlus does not attempt IPC until a numeric ID is supplied. Otherwise, confirm Discord Desktop is running in the same user/session, the numeric ID is correct, authorization was run with the temporary `LCDSIRPLUS_DISCORD_CLIENT_SECRET`, the current account was authorized, required tester access exists, the account is in a voice channel, and someone allowed by `discord_show_self` is speaking. Reauthorize after revocation or client-ID changes. Never share token files.

Application says it is already running

Only one normal/direct-HID runtime may own the current Windows session. Exit the existing tray application normally before starting another or a direct hardware test.

Logs, diagnostics, and privacy

The default log is `%LOCALAPPDATA%\LCDSirPlus\lcdsirplus.log`. Rotation is bounded by `log_max_bytes` and `log_backups`. `--diagnostic-dir PATH` selects a different log/diagnostic directory for that run. Normal tray launch has no console window. A configuration or logging failure during desktop startup is shown in a user-dismissible native error dialog; one-shot command modes report failures through standard error instead.

Create an offline diagnostics bundle with:

```
.\LCDSirPlus.exe --diagnostics  
.\LCDSirPlus.exe --diagnostics --diagnostic-dir C:\safe\local\directory
```

Diagnostics mode ignores `--config`, starts no hardware/providers/probes/ Discord/actions, and writes a store-only ZIP capped at 1 MiB. It contains only `privacy.txt`, `report.txt`, and `manifest.txt`. It excludes raw logs and configuration, credentials, Discord content and identifiers, process/window details, user paths, network targets, environment and registry values, serials, and device paths. Review any bundle before sharing. Raw logs and configuration may still contain operational detail and should not be posted without private review.

LCDSirPlus is local-first. Normal optional outbound traffic is limited to a configured network probe and Discord token exchange/refresh. LHM is restricted to loopback. HWINFO access is read-only shared memory, and PresentMon uses local ETW/captured frame output. No LCDSirPlus feature opens an inbound listener.

Normal launch is a GUI/tray process with no console. One-shot commands attach output to a parent console or redirected handles. Interactive PowerShell does not always wait for a GUI-subsystem executable before returning the prompt; use a pipeline such as `& .\LCDSirPlus.exe --help 2>&1 | Out-String` when output and synchronous completion matter, or use `Start-Process -Wait -PassThru` when only waiting/exit status is needed. A normal desktop startup failure such as invalid configuration or an unavailable log path is also shown in a dismissible **LCDSirPlus startup error** MessageBox; one-shot command failures stay on standard error.

Update and uninstall

For a published update or repair, close LCDSirPlus, verify the new setup EXE, and run it in the same install mode. Rerunning setup restores installer-owned files. Windows Modify, where offered, runs the exact setup cached at `installer\LCDSirPlus-Setup.exe` with the registered `/CURRENTUSER` or `/ALLUSERS` scope. Setup refuses an opposite-scope registration, an old PowerShell install in the selected directory, or a foreign startup-task collision when owned by the same account; remove that account's old installation first. An all-users registration owned by another account may coexist with this account's current-user install. Setup checks the active account and does not enumerate offline user hives. Installed configuration remains at

%LOCALAPPDATA%\LCDSirPlus\Config\lcdsirplus.txt ; logs and credentials remain elsewhere under %LOCALAPPDATA%\LCDSirPlus .

Uninstall from Windows **Installed apps**. The uninstaller removes application files, shortcuts, and the product-unique startup task. It preserves %LOCALAPPDATA%\LCDSirPlus . Task ownership and removal are checked before any application files are deleted; initialization only validates, while confirmed uninstall revalidates and deletes the task immediately before file removal. A cancel leaves the task intact, and a Task Scheduler failure leaves the installation available for retry. Delete user data manually only when its logs, configuration, and Discord credentials are no longer needed. Revoke Discord authorization separately if needed.

For portable removal, first set start_at_login 0 , start that exact portable executable once, and exit it so its owned Run value is removed. Then delete the portable directory. This cleanup never removes a foreign Run value.

Slot option reference

The following table is the complete set of 53 tokens accepted in slot_0 through slot_3 . The packaged modules.md is the shorter button-slot quick reference; this table remains standalone and includes each source, prerequisite, and fallback.

Exact token	Display	Source/config dependency	External dependency and fallback
HEADSET_BATTERY	SteelSeries wireless battery/charge/online state	Exact supported USB receiver HID profile; headset_enabled , headset_*	Supported Arctis/GameBuds receiver family; otherwise DISABLED , NO DONGLE , OFFLINE , or STALE .
CONTROLLER_BATTERY	Controller number, battery, or wired	XInput; controller_enabled , controller_index , poll setting	Connected XInput controller; otherwise OFFLINE .
FPS_CURRENT	Current FPS	PresentMon capture and presentmon_*	Bundled PresentMon and selected game; otherwise N/A / STALE .
FPS_1LOW	1% low FPS	PresentMon and presentmon_window_ms	Same PresentMon fallback.
FPS_01LOW	0.1% low FPS	PresentMon and presentmon_window_ms	Same PresentMon fallback.
FRAME_TIME	Frame time in ms and trailing 30-second graph	PresentMon	Same PresentMon fallback.
CPU_TEMP	CPU package temperature	Exact HWiNFO pair, then LHM loopback	No generic Windows source; N/A / STALE without a usable fallback.
GPU_TEMP	GPU temperature	NVAPI/ADLX; LHM temperature fallback	Vendor driver or LHM; otherwise N/A / STALE .
NET_IN	Instantaneous ingress SI bit rate	Native interface counters, one-second sample	No external program; N/A in safe mode/provider failure.
NET_OUT	Instantaneous egress SI bit rate	Native interface counters	Same as NET_IN .
NET_BOTH	Ingress plus egress SI bit rate	Both native direction readings	N/A / STALE unless both are current.
NET_IN_GRAPH	Ingress over exactly the trailing 30 seconds	Native counters; graph ceiling setting	Empty/ N/A without throughput.
NET_OUT_GRAPH	Egress over exactly the trailing 30 seconds	Native counters; graph ceiling setting	Empty/ N/A without throughput.
NET_GRAPH	30-second graph, ingress above/egress below center	Both native directions; graph ceiling setting	N/A if either direction is unavailable or nonfinite; STALE if either valid direction is stale.

Exact token	Display	Source/config dependency	External dependency and fallback
PING	Latest latency in ms	Enabled network probe and target/method/timing settings	Sends configured ICMP/TCP; N/A when disabled/lost, stale on provider error.
JITTER	Latency variation in ms	Probe window	Requires at least two successful probes.
PACKET_LOSS	Probe loss percentage	Probe window	N/A without enabled/history data; stale on provider error.
MIC_STATUS	LIVE / MUTED	Core Audio; audio_enabled , poll setting	Readable default microphone; otherwise N/A .
AUDIO	Output volume percent	Core Audio; audio_enabled , poll setting	Readable default output; otherwise N/A .
SESSION_TIME	Active capture elapsed time	PresentMon session	PresentMon/active game; 00:00 when inactive.
SESSION_SUMMARY	Fitted duration and stutter count, or IDLE	PresentMon and stutter threshold	PresentMon/active game; otherwise IDLE .
CLOCK	Local time	Windows clock; accepted time_format is currently a no-op	None.
GAME_NAME	Captured game/process name	PresentMon target	N/A without a selected name.
ALERTS	Alert count and spaced level (n L 2 / n L 3), or CLEAR	Alert thresholds and corresponding current telemetry	Missing telemetry creates no false alert; physical/preview slot 4 acknowledges globally.
PROVIDER_STATUS	Tracked providers down or OK	Runtime provider health	Safe mode publishes only healthy safe-mode and therefore displays OK .
CPU_LOAD	Whole-system CPU percent	Native scheduler accounting	None.
RAM_USAGE	Percent and used/total RAM	Native Windows memory status	No external program.
GPU_LOAD	GPU utilization percent	NVAPI/ADLX and gpu_provider	Vendor driver; N/A / STALE if unavailable. No LHM fallback.
VRAM_USAGE	VRAM percent and used/total	NVAPI/ADLX and gpu_provider	Vendor memory telemetry; N/A / STALE if incomplete. No LHM fallback.
CPU_CACHE_TEMP	Intended Cache-domain CPU temperature	Requires one Cache domain descriptor and temperature	Currently no runtime producer, including LHM; always N/A .
CPU_FREQ_TEMP	Intended Frequency-domain CPU temperature	Requires one Frequency domain descriptor and temperature	Currently no runtime producer, including LHM; always N/A .
CPU_LOAD_GRAPH	CPU load and trailing 30-second graph	Native scheduler accounting; fixed 0..100% scale	No external program.
GPU_LOAD_GRAPH	GPU load and trailing 30-second graph	NVAPI/ADLX; fixed 0..100% scale	Vendor driver; no LHM load fallback.
CPU_TEMP_GRAPH	CPU temperature and trailing 30-second graph	CPU source; cpu_temp_max_c ceiling/warning threshold	Same fallback as CPU_TEMP .
GPU_TEMP_GRAPH	GPU temperature and trailing 30-second graph	GPU source; gpu_temp_max_c ceiling/warning threshold	Same fallback as GPU_TEMP .
VRM_TEMP	VRM temperature	Exact 1hm_vrm_temp_sensor	User-enabled LHM loopback only.
CPU_FAN	CPU-fan percentage	Exact LHM duty, else RPM / configured max	cpu_fan_max_rpm 0 disables RPM estimate.
PUMP_RPM	Pump percentage despite compatibility name	Exact LHM duty, else RPM / configured max	pump_max_rpm 0 disables RPM estimate.

Exact token	Display	Source/config dependency	External dependency and fallback
POWER_LIMIT	Authoritative total power	Exact HwINFO total pair, then exact LHM total sensor	Never synthesized; N/A without exact total.
CPU_GPU_POWER	Complete CPU+GPU subtotal	Current CPU and GPU component power	N/A if either component is absent/stale; not total system power.
CHIPSET_TEMP	Chipset temperature	Exact LHM SensorId	User-enabled LHM loopback only.
MOTHERBOARD_TEMP	Motherboard temperature	Exact LHM SensorId	User-enabled LHM loopback only.
DISK_IO	Aggregate system read/write rates	Native PDH PhysicalDisk(_Total)	No external program.
DISK_IO_GRAPH	Compact spaced read/write labels and trailing 30-second graph	Native PDH; disk_graph_ceiling_mbps	Scale clips graph height; DISK_IO retains fuller rates.
RAM_DETAIL	Fitted used/total physical RAM in IEC units	Native Windows memory status	No external program.
FPS_GRAPH	FPS and trailing 30-second graph	Bundled PresentMon; fps_graph_ceiling	Active capture required.
THERMALS	Complete CPU nC and GPU nC rows	Both current temperature readings	Either threshold can invert this pane.
CONNECTIONS	Established TCP count	Native IPv4+IPv6 TCP tables	No external program.
NET_HEALTH	Three rows: P nMS, J nMS, and L n%	Enabled probe and target/settings	All three readings required; blank scanlines separate rows.
SYSTEM_BATTERY	AC/battery/charging state	Native Windows system power status	Shows AC/BAT/CHG/no-battery/unknown/stale states.
HARD_FAULTS	Approximate page-read pressure as n/s	Native PDH Page Reads/sec	Titled FAULTS; not an exact hard-fault count.
BOTTLENECK	CPU/GPU/RAM/DISK I/O/NONE heuristic	Native metrics, thresholds, sustain	NONE dynamically displays next token without changing selection.
PROC_HANG	Selected hung window and guarded emergency hold action	Native detector and selected-slot physical button; hang_enabled 0 by default	Unique across slots; provider disabled/unavailable and no-target fall-through until opt-in, then short release navigates and the hold threshold acts automatically; physical disposable-child acceptance remains required.

For exhaustive configuration key defaults and validation ranges, see CONFIGURATION.md. For physical action acceptance, see HARDWARE-ACCEPTANCE.md.